

Supplementary Material

Which Hand-Motion Composition Axes Are Worth Measuring.
Section numbers in roman refer to the main paper.

S1. Anatomical projection and the usable-DOF audit

We count usable DOF out of 27 before and after the per-joint convention inference of Section III-A. Dominance is the flexion axis's share of joint variance, and the residual is the mean per-joint angular error, in degrees, of reprojecting the anatomical DOF back to MANO's axis-angle parameters. The count exempts the three wrist-rotation channels, so 26 of 27 means 23 of the 24 that MANO can express. That is not a defect, for the reason Section III-A gives; filling those channels would encode where the camera stood.

TABLE S1: usable degrees of freedom before and after the fix

dataset	usable DOF	dominance	residual
OakInk-Image	23 → 26	0.663 → 0.726	6.01°
OakInk2	19 → 23	0.470 → 0.571	21.86° → 10.22°
DexYCB	19 → 23	0.470 → 0.537	12.26°
GRAB	16 → 21	0.450 → 0.533	16.15°

Provenance is uneven, and we record it rather than smoothing it. No released artifact carries a before usable-DOF count for any dataset, so those four figures come from the project's working notes. OakInk2's and GRAB's before dominances are in the rebuild logs. OakInk-Image's and DexYCB's come from the same notes and are likewise unreleased.

S2. Parameter counts, realized rate and the KL floor

Section V-A's sweeps use $K = 12$, where the baseline has 2,283,059 parameters and the modular arm 2,295,727, a difference of 0.6%. The pooled banks of Sections V-C and V-D use $K = 16$ (2,297,907), and Section V-C's two weight sweeps use $K = 12$.

Rates below are final-epoch validation quantities in nats per frame, with the assignment rate estimated as usage entropy minus assignment entropy. The baseline column is the continuous latent and the modular column the assignment channel, so the ratio is modular over baseline throughout.

TABLE S2: realized information rate, baseline against modular

sweep	n models	baseline rate	modular assignment rate	ratio
confirmatory category axis, original sweep	140 / 139	0.207	2.275	11.0×
pc_hard, the positive control, both budgets	46 / 46	0.223	2.399	10.8×
pc_easy at	10 / 10	0.230	2.424	10.6×

budget 256, the negative control				
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The baseline's KL penalty is active and binding rather than a formality. The prior implementation clamps per dimension, and the clamped KL exceeds its $12 \times 0.02 = 0.24$ floor on the training split in 102 of 140 baselines on the confirmatory axis, 45 of 46 on the positive control and 19 of 20 on the interpolable control. That last figure is taken over both pc_easy budgets rather than the budget-256 row of Table S2. The rate-control re-sweep is weaker on both counts, at 92 of 140 and a per-model minimum of zero active units, with three unmatched baselines fully collapsed; excluding the six seeds below 0.9 active units moves its unmatched advantage from +0.00693 to +0.00649. Nor is the latent collapsed, with mean active units between 0.996 and 1.000 across the three sweeps and a per-model minimum of 0.833. On the other side, a conditional assignment entropy of 0.160 nats against a $\ln 12 = 2.485$ maximum means the selection is nearly deterministic given the input, which is what makes the assignment a high-rate channel.

S3. Gates, coverage and the breadth rule

Table I of the main paper names the five gates. The table below gives what each caught in full, including the figures the released gate now reports where it still exits non-zero.

TABLE S3: the gate transcripts in full

Gate	Caught
primitive collapse	<code>primitives_used = 1.0</code> , a "modular" model that is monolithic in disguise. No standalone artifact is released. Over every $K = 12$ modular model in the sweeps Table II reports, the minimum final-epoch <code>val_primitives_used</code> is 9.732 of 12 and exactly 16 sit below 11.0, all of them OakInk2's; the withheld fine object $\times$ intent sweep adds a seventeenth at 10.0. The $K = 16$ banks of Section V-D and of Section V-C's pooled arm are outside that scope
informed coverage	held compositions absent from the informed set, which pins the penalty at zero by construction. The round-robin swap covers 5 of 5 on 18 of 20 confirmatory seeds and 16 of 20 affordance seeds, so the gate exits 1 on both
composition leak	48–69% of targets had their exact fine composition in the informed training set against 0% for naive, pre-fix; the released gate now reports 0.0% on both arms
DOF health	<code>thumb_mcp_abd</code> pinned at its limit in 100% of GRAB and DexYCB frames under an assumed abduction axis (pre-fix; the released gate now reports 40% and 33%, still above its own 25% threshold, so it exits 1; see Section S5)
split balance	the risk that an unpaired design measures set difficulty. The gate tests the mean imbalance and passes at +1.4%; the per-seed spread it does not test is −6.1% to +10.7% (Section III-C)

Not every sweep is gated per seed. The class sweep, the rate-control sweep on the interpolable control and the 17 completing seeds of the confirmatory axis record a soundness block for each seed. The confirmatory axis's first 52 predate that record, so standalone runs of the coverage and leak gates over a seed subset cover them instead. The split is deterministic given the seed, so those runs bound the same quantity. The coverage gate covers 37 of the 69, 20 as a standalone run and 17 through per-seed soundness blocks, which leaves seeds 20 to 51 unrecorded. The leak gate covers all 70, both as a standalone artifact and in the per-seed soundness blocks of the rate-control sweep, which report 0.0% on both arms throughout.

Breadth is the fraction of held compositions the informed arm reaches, and a row passes at 0.8. Four of GRAB's 80 rows reach 5 of 6 and pass on that rule, where a row reaching 4 of 5 scores exactly 0.80 and fails. Under it the class axis fails 6 of 24 rows, with seeds 1, 5 and 6 reaching 3 of 4 held compositions, and the three larger axes fail a minority: 4 of 40 confirmatory, 8 of 40 affordance, 4 of 40 id-prefix. The seeds excluded from the confirmatory sensitivity check of Section V-B are those flagged by the standalone gate, seeds 8 and 13, together with seed 59, flagged by a per-seed soundness block.

S4. Sensitivity checks, control floors and withheld sweeps

Every headline in Section V is bounded by a control that should return zero. On the raw penalty the unmatched arms differ by +0.00154 at budget 256 on the interpolable control, and by +0.00155 against the rate-matched baseline, which is 61% of the rate-matched effect of +0.00254. On the normalized penalty the same control returns +0.018 unmatched and +0.099 matched, the latter at 18 of 20 and clearing both declared tests, which is 90% of the +0.111 the real axis returns against the same arm. The baseline arm's own difficulty on that control is +0.00258 at budget 256 and +0.00306 at budget 64, against a re-sweep difficulty of +0.01024 unmatched and +0.00586 matched.

The normalized metric fails as a robustness check because its maximum sits away from the claim. The deprecated id-prefix axis gives +0.155 at 53 of 70, which is real in that the axis has both difficulty and the largest raw advantage. But GRAB's shape axis carries no measurable difficulty and still returns +0.116 at 25 of 40 ($t p = 0.035$) against its unmatched baseline, above the +0.104 the confirmatory axis returns against the same arm, while the withdrawn DexYCB axis returns +0.043 against the confirmatory +0.046. Real data therefore bounds the metric harder than the synthetic control does.

Two exclusions test the rate-matched intervention itself. Five of its 70 seeds cover only 4 of 5 held compositions, and excluding them moves the raw figure to +0.00273 at $p = 0.052$, so they carry neither reading. On the confirmatory axis, excluding the three flagged seeds of Section S3 moves the advantage from +0.00397 over 69 seeds to +0.00419 over 66 ($t p = 0.0010$). On the class axis, three of twelve seeds cover 3 of 4 held compositions, which is why Section V-A reads its sign test conservatively.

The three off-the-shelf decompositions of Section V-A are read against baseline penalties of +0.00904 for the bank sweep, +0.01018 for field and +0.00871 for MANN, in that same order. The three pairwise contrasts between the arms themselves are released and not argued from, as none separates at $n = 8$ and we ran no equivalence test.

Two candidate explanations for the interaction fail, and Section V-B reports only their outcome. The first holds that OakInk-Image's intents are more pose-relevant, and its own numbers refuse it, as the intent main effect is 0.045 there and 0.044 on GRAB. The second holds that a crossed design produces it by recording one object under several named intents, and it fails harder. Over the eleven screened axes, grid occupancy and interaction excess correlate at $r = -0.095$ ($p = 0.78$); GRAB's object $\times$ intent-class grid is 93% occupied with a significantly negative excess; both DexYCB axes are perfectly crossed and carry the two most negative scores; and OakInk-Image's 86%-crossed fine axis is still unsplitable.

Two sweeps are withheld, and neither supports a conclusion. DexYCB's subject $\times$ shape sweep fails the coverage gate on all twelve seeds at both budgets, the informed arm reaching a mean of 1.58 of 4 held compositions. The fine object $\times$ intent sweep of OakInk-Image runs five seeds over 345 occupied cells at 2.23 clips each and returns -0.00163 at budget 256 and -0.00343 at budget 64, both negative and both artifacts of grid thinness rather than evidence against a modular advantage.

S5. Transfer detail and five further limitations

The equivalence test of Section V-D needs more clips than these banks supply. At the pooled spread of 2.40° and a 0.36° difference over 165 clips per arm, a one-degree margin needs 176 per arm for 80% power and 243 for 90% at $\alpha = 0.05$ by the normal-approximation TOST formula. On the 105 clips left after the degenerate block is dropped, spread 2.70° and difference 0.57° , it needs 488. Each bank's seen arm is bounded by the primitive pairs its training data contains, at 19, 23, 23, 40 and 72. The last contributed 60 of its 72, so the 165 clips are $19 + 23 + 23 + 40 + 60$ and only four bounds bind. More clips therefore means more banks, not more sampling.

The retargeting figures of Section V-D are read per dataset, in the order OakInk-Image, DexYCB, GRAB throughout. Direct retargeting of the first 40 trajectories per dataset clips 0.09%, 4.68% and 10.20% of DOF values and interpenetrates in 24.6%, 76.8% and 59.0% of frames. The abduction correction resolves 92%, 97% and 64% of that at a cost of 0.90°, 5.48° and 5.46° of deviation, leaving 2.1%, 2.1% and 21.2% of frames penetrating. GRAB is the residual case in every column. Two 160-clip task-comparison files in the release are void rather than negative, their monolithic arms having trained 10 to 13 epochs on a window-level latent.

Five further limitations are documented with the release, each with its own artifact. First, the advantage does not grow as data shrinks, over four sweeps at budgets 64 and 256 with two reversing and none significant. Second, contact does not explain the nulls, as adding GRAB's 16 per-region contact channels leaves the penalty at +0.00032 ($p = 0.813$). Third, abduction is clipped everywhere, leaving 1 to 6 of 27 DOF pinned, and wrist rotation is unrepresented for the reason Section S1 gives. Fourth, both synthetic controls predate the split fixes, so their partitions are unreproducible even though their headline figures are stable. Fifth, the interpenetration figure recorded in the composition metadata is a capsule proxy that does not track the mesh test even in rank, so only the mesh figures are quoted in the main paper.